ARALYTICS

Feature test report

Live deployment review
aralytics.cleolyn.dev

31 October 2026 | Asia/Manila
Student, teacher, API, storage, reports and access controls

Overall result: works in many core flows, with fixes needed

The tested deployment supports enrollment, lessons, quizzes, task submission and grading, student progress, and report exports. Two confirmed defects remain: lesson attachment to interventions fails, and an older student dashboard API returns a database error. The current student dashboard screen itself loads successfully.

Live API and data assertions	Passed	Failed
132	129	3

Counts represent individual checks, not a percentage of all possible features. Three failed assertions correspond to two underlying defects. Browser observations and local regression suites are reported separately.

Most important findings

Fix first: teachers cannot attach a newly uploaded lesson to a remediation plan. Creating a plan with that lesson or modifying a plan returns HTTP 500.

Fix next: GET /api/student/dashboard refers to columns that do not exist in the deployed quiz table. The active dashboard screen uses other working endpoints.

Coverage gap: the teacher quiz editor offers three choice-based styles. The backend accepts six question formats, so true/false, identification and matching cannot currently be authored through that screen.

This was an audit, not a repair deployment. Findings below describe observed behavior; no application code fixes were deployed during testing.

What works - feature coverage

Feature area	Result	Evidence and practical limit
Site, HTTPS and SPA routes	PASS	Homepage, login, registration, help and deep links return 200. Browser pages render.
Email sign-in and sign-out	PASS	Real Clerk UI sign-in for QA student and teacher; verification-code flow and return to login on sign-out. Test emails were used.
Class invitations and enrollment	PASS	Create/validate code, reject invalid LRN, request, pending gate, duplicate rejection, adviser approval and code closure.
Teacher and student access	PASS	Signed-out requests rejected; student cannot use teacher APIs or exports; teacher cannot use student API. Malformed token rejected.
Lessons and D1 uploads	PASS	Upload, edit, list, detail and download. Uploaded bytes match downloaded bytes. Valid PDF opens in lesson viewer.
Reading and offline-save controls	PARTIAL	Reading logs sync; progress displays read lesson. Save offline shows Saved offline. Actual network-disconnected relaunch was not exercised.
Quiz creation and drafts	PASS / LIMITED	API publishes six formats. Teacher UI draft survives reload and publishes a choice-based quiz. Only three styles are offered in the editor.
Quiz taking and grading	PASS	Legacy six-format submission scored 6/6. Current QuizRunner API accepts answers, grades on server, rejects duplicates and ignores a tampered client score. Browser-authored quiz was completed through the real student screen with 1/1.
Screen-monitor telemetry	PASS / PARTIAL	Proctor event and current security-log endpoint accept QA events; teacher audit loads. Full timed/fullscreen behavior was not stress tested.
Performance tasks	PASS	Create task; PDF submission; teacher download/queue/grade; student sees 90/100 and feedback; peer downloads blocked.
Upload and score validation	PASS	Unsupported extension, invalid PDF header, over-5-MB file and grade above total rejected.
Progress and answer review	PASS	Browser shows quiz 6/6 and task 90/100. See answers displays six correct answer records.
Analytics and reports	PASS	Teacher overview, classes, gradebook, profiling, queue and item analysis return data. PDF/XLSX exports contain populated assessment data.
Practice/interventions	PARTIAL / FAIL	Failed quiz creates intervention; approve, override and student practice display work. Linking an uploaded lesson fails.

PASS means the stated scenario succeeded in this run. It does not guarantee every edge case or every supported device.

Confirmed defects and recommended fixes

D-01 | Uploaded lesson cannot be attached to an intervention

Priority: high. This blocks a teacher from assigning an uploaded review resource through Student support.

Reproduction: upload a lesson using the teacher materials flow; use its returned material ID as material_id when creating an intervention, or when modifying an existing intervention.

Observed: POST /api/analytics/interventions returned 500 (expected 201). POST /api/teacher/intervention/:id/action with MODIFY returned 500 (expected 200). Both returned: SQLITE_CONSTRAINT: SQLite error: FOREIGN KEY constraint failed.

Diagnosis: lesson uploads are saved in instructional_materials, while intervention recommended_material_id references materials. The uploaded lesson ID is missing from the table expected by the intervention relation. This is supported by the route/controller definitions and the two live failures.

Recommendation: unify the resource identifier used by uploads, teacher selectors and intervention foreign keys, or provide an explicit mapping. Retest create, modify, download and student practice after migration.

D-02 | Older student dashboard endpoint returns 500

Priority: medium. GET /api/student/dashboard returned 500 for an approved QA student. Error: no such column: quarter. Schema inspection also found description and is_active missing from the quiz table, although the query requests them.

Visible impact: the current student dashboard screen loaded successfully using the separate learner-data endpoints. This is a broken API, not a demonstrated failure of that screen.

Recommendation: align the query with the deployed schema, or remove the unused route. Add an approved-student API check to release verification.

G-01 | Question-format authoring gap

Priority: medium if all six formats are required. The editor offers Standard question, Real-life situation and Reading comprehension. It requires four choices for every item. True/false, identification and matching are accepted by backend grading but are absent from the editor.

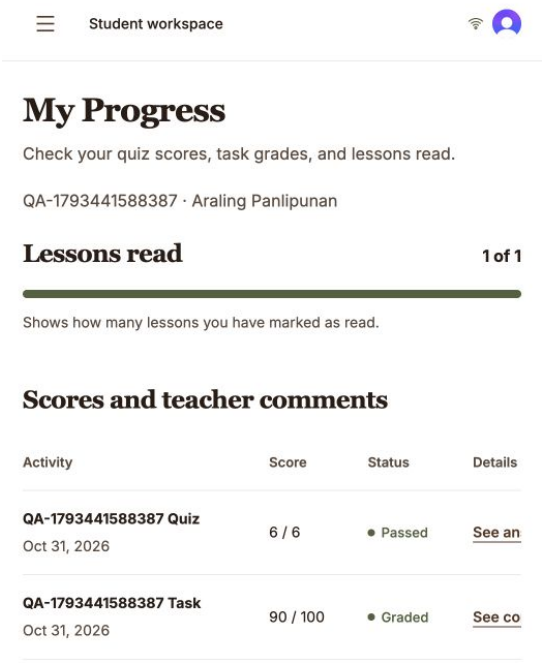
Recommendation: decide whether the product supports three styles or six formats, then make the editor, requirements and test suite agree.

Additional observations

Uploaded lesson URLs were downloadable without authentication. This supports the current offline-reading design, but should be checked against the intended class privacy policy. Submission files were protected. Server errors also expose database details; use generic user-facing errors and keep diagnostics in server logs.

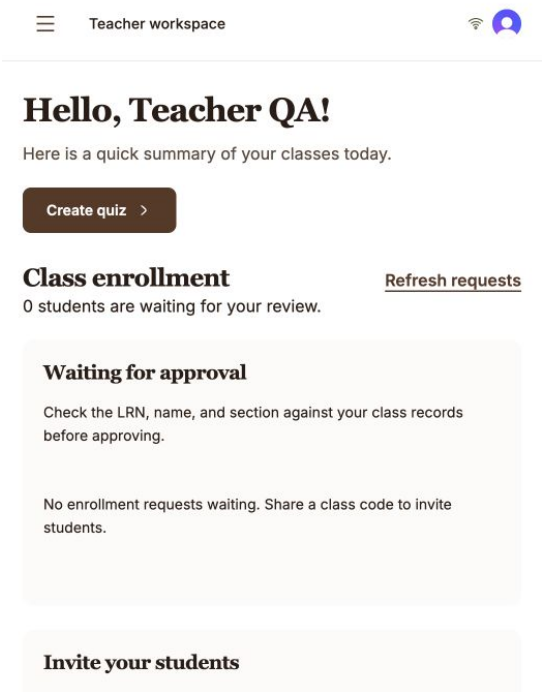
Browser evidence from synthetic QA accounts

Student progress: persisted grades and feedback access



The student page displays the six-format quiz score of 6/6 and the graded task score of 90/100. All displayed data belongs to this QA run.

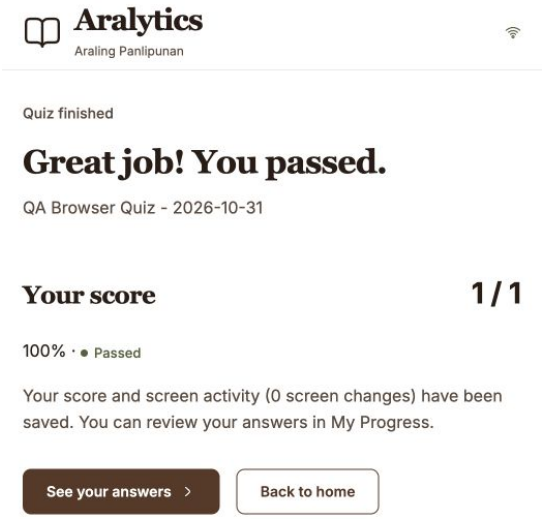
Teacher dashboard: enrollment and class activity



The teacher workspace loads with the two approved QA learners, published quizzes, and a pending practice recommendation.

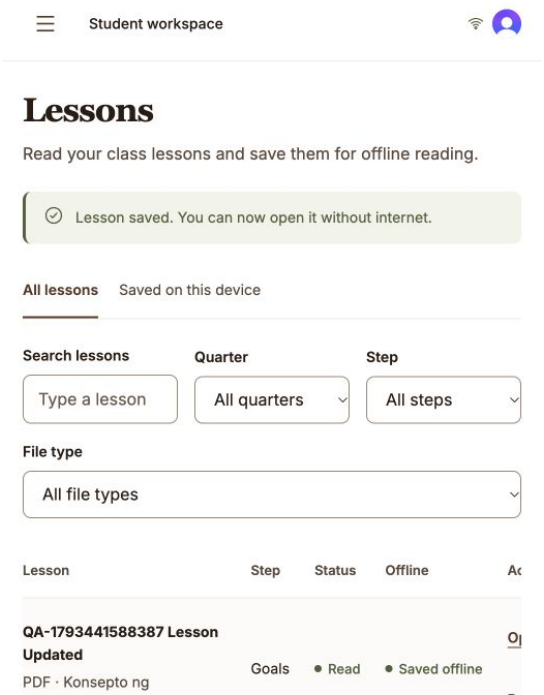
End-to-end quiz and offline-save evidence

Teacher-authored quiz completed by a student



The quiz was drafted, reloaded and published through the teacher UI. A student selected an answer, marked it for review, submitted through the real QuizRunner screen and received 1/1.

Offline-save control confirms local availability



Save offline completed and the library showed Saved offline. This confirms the save flow, not a full disconnected-browser restart.

Regression results, limits and follow-up

Local regression suites

The repository tests were run against disposable SQLite databases, then rerun with local sample fixtures. No regression suite was pointed at the live Turso database. After fixture setup, 8 of 11 server suites passed. The separate client authentication-flow suite also passed.

Suite	Result	Interpretation
analytics_engine	PASS	Passed in a disposable seeded database.
auth_multi_user	FAIL	Old JWT-based middleware expectation conflicts with current Clerk identity verification.
enrollment_flow	PASS	Passed in a disposable seeded database.
master_loop_modules	FAIL	Mock upload requests omit req.app; storage helper expects Express app.locals. Live uploads passed.
offline_policy	PASS	Passed in a disposable seeded database.
persistence_module	PASS	Passed in a disposable seeded database.
rbac_security	PASS	Passed in a disposable seeded database.
redesign_flow	PASS	Passed in a disposable seeded database.
srs_compliance	FAIL	Question-count assertion depends on richer seeded fixtures. Live six-format grading and answer review passed.
upload_storage	PASS	Passed in a disposable seeded database.
verification_gate	PASS	Passed in a disposable seeded database.

Not fully verified

Google OAuth completion; delivery to a real email inbox; new-user verification with real mail; password recovery; account deletion; production Clerk configuration; cross-teacher data isolation; prolonged timer expiry; concurrent classroom load; live Free-plan CPU/quota exhaustion; full offline cold start and reconnect; broad mobile and cross-browser accessibility testing.

The supplied Clerk instance remains in development mode. Test email verification used Clerk's documented +clerk_test flow. No real email delivery was assumed from those tests.

Suggested order of work

1. Fix intervention resource references and retest the complete practice-resource assignment.
2. Correct or retire the broken dashboard API.
3. Resolve the authoring-format gap and update legacy tests.
4. Complete real-email/Google, offline recovery, privacy and load checks before broad classroom rollout.

Test method and evidence notes

Environment

Live target: https://aralytics.cleolyn.dev
Frontend: React/Vite SPA on Cloudflare Static Assets.
API: Express on Cloudflare Workers. Application database: Turso.
Upload storage: Cloudflare D1. Authentication: Clerk development instance.

Data safety

Three isolated QA identities and labeled QA content were used. Cleanup removed the 3 Clerk identities and their application records, plus 3 QA upload objects. Verification found 0 QA application users remaining. Existing user accounts and the six pre-existing uploaded files were not targeted.

How evidence was checked

Live requests checked status codes, expected role denials and response content. Upload tests compared downloaded bytes with uploaded bytes. Scores were read back from persisted results. Excel workbooks were parsed and found to contain populated assessment rows; PDF exports were parsed as PDF documents with text. Browser tests verified visible UI states, rather than treating HTTP 200 as proof of a working screen.

Important distinctions

The six-format grading check used the legacy submission route and scored 6/6; the current assessment route was also tested for successful submission, server-side grading, duplicate protection and remediation. A passing file signature alone was not used as the final export-content check. A failed mock or missing fixture was not treated as a confirmed live product failure.

Reproducibility

QA run identifier: QA-1793441588387. Test date: 31 October 2026, Asia/Manila. The appendices list live assertions and their observed HTTP results. Internal evidence includes API-result JSON, seeded regression logs and UI screenshots. Credentials, session tokens and real user details are excluded from this report.

Reference for the test verification method

Clerk documentation: test emails and verification codes

Report scope

This is a broad functional assessment of the identified feature areas, not a proof that every possible input, browser state, attack path or production load works. Items not exercised are explicitly listed above. No application repairs were bundled into this audit.

Appendix - live API and data assertions

132 assertions: 129 passed and 3 failed. The same endpoints may appear several times for different roles, payloads or persisted-data checks. Timing values are single-run observations, not a load benchmark.

# / Result	Check	Observed evidence
001 PASS	Public availability /api/health	HTTP 200; expected 200; 1206 ms
002 PASS	Public availability /api/test-db	HTTP 200; expected 200; 726 ms
003 PASS	Public availability /	HTTP 200; expected 200; 152 ms
004 PASS	Public availability /login	HTTP 200; expected 200; 93 ms
005 PASS	Public availability /register?role=STUDENT	HTTP 200; expected 200; 150 ms
006 PASS	Public availability /help	HTTP 200; expected 200; 154 ms
007 PASS	Public availability /student/dashboard	HTTP 200; expected 200; 127 ms
008 PASS	Public availability /teacher/reports	HTTP 200; expected 200; 110 ms
009 PASS	Public availability /manifest.json	HTTP 200; expected 200; 249 ms
010 PASS	Public availability /sw.js	HTTP 200; expected 200; 173 ms
011 PASS	Signed-out protection /api/auth/me	HTTP 401; expected 401; 79 ms
012 PASS	Signed-out protection /api/student/dashboard	HTTP 401; expected 401; 79 ms
013 PASS	Signed-out protection /api/interventions	HTTP 401; expected 401; 89 ms
014 PASS	Signed-out protection /api/teacher/classes	HTTP 401; expected 401; 90 ms
015 PASS	Signed-out protection /api/quizzes	HTTP 401; expected 401; 113 ms
016 PASS	Signed-out protection /api/student/scores	HTTP 401; expected 401; 116 ms
017 PASS	Signed-out protection /api/materials	HTTP 401; expected 401; 152 ms
018 PASS	Signed-out protection /api/analytics/proficiency	HTTP 401; expected 401; 980 ms
019 PASS	Signed-out protection /api/student/lessons	HTTP 401; expected 401; 1031 ms
020 PASS	Signed-out protection /api/export/pdf/report	HTTP 401; expected 401; 1157 ms
021 PASS	Signed-out protection /api/performance-tasks	HTTP 401; expected 401; 1179 ms
022 PASS	Signed-out protection /api/teacher/dashboard	HTTP 401; expected 401; 1182 ms
023 PASS	Signed-out protection /api/teacher/gradebook	HTTP 401; expected 401; 1332 ms
024 PASS	Signed-out protection /api/enrollment/status	HTTP 401; expected 401; 1487 ms
025 PASS	Profile sync TEACHER	HTTP 200; expected 200; 1207 ms
026 PASS	Profile sync STUDENT	HTTP 200; expected 200; 1059 ms
027 PASS	Profile sync STUDENT_PEER	HTTP 200; expected 200; 855 ms

# / Result	Check	Observed evidence
028 PASS	Unenrolled student blocked	HTTP 403; expected 403; 628 ms
029 PASS	Student cannot create class invitations	HTTP 403; expected 403; 191 ms
030 PASS	Teacher creates invitation	HTTP 201; expected 201; 536 ms
031 PASS	Reject invalid class code	HTTP 400; expected 400; 155 ms
032 PASS	Validate class code	HTTP 200; expected 200; 546 ms
033 PASS	Reject invalid LRN	HTTP 400; expected 400; 550 ms
034 PASS	Student requests enrollment	HTTP 201; expected 201; 975 ms
035 PASS	Pending student blocked	HTTP 403; expected 403; 633 ms
036 PASS	Reject duplicate enrollment	HTTP 409; expected 409; 338 ms
037 PASS	Teacher reviews enrollment queue	HTTP 200; expected 200; 922 ms
038 PASS	Teacher approves student	HTTP 200; expected 200; 614 ms
039 PASS	Peer requests enrollment	HTTP 201; expected 201; 1024 ms
040 PASS	Teacher approves peer	HTTP 200; expected 200; 574 ms
041 PASS	Student cannot enter teacher API	HTTP 403; expected 403; 241 ms
042 PASS	Teacher cannot enter student API	HTTP 403; expected 403; 205 ms
043 PASS	Student lessons	HTTP 200; expected 200; 353 ms
044 PASS	Authenticated read /api/interventions/flagged	HTTP 200; expected 200; 507 ms
045 PASS	Authenticated read /api/quizzes	HTTP 200; expected 200; 554 ms
046 PASS	Teacher classes	HTTP 200; expected 200; 613 ms
047 PASS	Student scores	HTTP 200; expected 200; 615 ms
048 FAIL	Student dashboard	HTTP 500; expected 200; 617 ms; SQL_INPUT_ERROR: SQLite input error: no such column: quarter (at offset 11)
049 PASS	Authenticated read /api/analytics/remediation-queue	HTTP 200; expected 200; 704 ms
050 PASS	Teacher profiling	HTTP 200; expected 200; 844 ms
051 PASS	Authenticated read /api/analytics/proficiency	HTTP 200; expected 200; 867 ms
052 PASS	Student practice	HTTP 200; expected 200; 901 ms
053 PASS	Authenticated read /api/interventions	HTTP 200; expected 200; 1625 ms
054 PASS	Teacher gradebook	HTTP 200; expected 200; 1627 ms
055 PASS	Teacher dashboard	HTTP 200; expected 200; 1630 ms
056 PASS	Authenticated read /api/performance-tasks	HTTP 200; expected 200; 1668 ms

# / Result	Check	Observed evidence
057 PASS	Authenticated read /api/materials	HTTP 200; expected 200; 1917 ms
058 PASS	Student interventions	HTTP 200; expected 200; 1944 ms
059 PASS	Teacher remediation-queue	HTTP 200; expected 200; 2250 ms
060 PASS	Curriculum taxonomy	HTTP 200; expected 200; 447 ms
061 PASS	ILAW curriculum filter	HTTP 200; expected 200; 627 ms
062 PASS	Reject invalid ILAW filter	HTTP 400; expected 400; 275 ms
063 PASS	Teacher uploads lesson to D1	HTTP 201; expected 201; 663 ms
064 PASS	Uploaded lesson byte integrity	HTTP 200; expected 200; 99 ms
065 PASS	Lesson public download behavior	HTTP 200; expected 200; 89 ms
066 PASS	Lesson detail	HTTP 200; expected 200; 594 ms
067 PASS	Lesson edit	HTTP 200; expected 200; 423 ms
068 PASS	Offline reading log sync	HTTP 200; expected 200; 942 ms
069 PASS	Reading log retrieval	HTTP 200; expected 200; 484 ms
070 PASS	Teacher authors six-format quiz	HTTP 201; expected 201; 967 ms
071 PASS	Student reads quiz without answer key	HTTP 200; expected 200; 502 ms
072 PASS	Student starts quiz	HTTP 201; expected 201; 694 ms
073 PASS	Peer cannot read another submission	HTTP 403; expected 403; 667 ms
074 PASS	Proctoring event	HTTP 200; expected 200; 1078 ms
075 PASS	Teacher proctoring audit	HTTP 200; expected 200; 234 ms
076 PASS	Student submits six-format quiz	HTTP 200; expected 200; 1689 ms
077 PASS	Six-format quiz score equals full marks	score: 6; note: Verified using persisted result; initial response reader used the wrong nested key.
078 PASS	Duplicate quiz submission rejected	HTTP 400; expected 400; 363 ms
079 PASS	Student sees own result	HTTP 200; expected 200; 436 ms
080 PASS	Teacher item analysis	HTTP 200; expected 200; 472 ms
081 PASS	Teacher creates performance task	HTTP 201; expected 201; 368 ms
082 PASS	Student reads task	HTTP 200; expected 200; 474 ms
083 PASS	Reject unsupported upload extension	HTTP 400; expected 400; 689 ms
084 PASS	Student submits PDF to D1	HTTP 200; expected 200; 657 ms
085 PASS	Submission download byte integrity	HTTP 200; expected 200; 498 ms

# / Result	Check	Observed evidence
086 PASS	Peer cannot download another submission	HTTP 404; expected 404; 372 ms
087 PASS	Anonymous cannot download submission	HTTP 401; expected 401; 103 ms
088 PASS	Teacher downloads submission	HTTP 200; expected 200; 298 ms
089 PASS	Teacher submission queue	HTTP 200; expected 200; 260 ms
090 PASS	Teacher grades task	HTTP 200; expected 200; 525 ms
091 PASS	Student cannot grade task	HTTP 403; expected 403; 278 ms
092 PASS	Student sees graded task	HTTP 200; expected 200; 445 ms
093 PASS	Teacher XLSX report export	HTTP 200; expected 200; 302 ms
094 PASS	Student blocked from xlsx export	HTTP 403; expected 403; 553 ms
095 PASS	Teacher PDF report export	HTTP 200; expected 200; 293 ms
096 PASS	Student blocked from pdf export	HTTP 403; expected 403; 273 ms
097 PASS	Close QA class invitation	HTTP 200; expected 200; 361 ms
098 PASS	Closed class code rejected	HTTP 400; expected 400; 333 ms
099 PASS	Verify legacy six-format score from persisted result	HTTP 200; expected 200; 2001 ms
100 PASS	Read authored quiz for active UI submission	HTTP 200; expected 200; 799 ms
101 PASS	Current UI quiz submission endpoint	HTTP 201; expected 201; 1454 ms
102 PASS	Current UI security telemetry endpoint	HTTP 200; expected 200; 250 ms
103 PASS	Create manual intervention	HTTP 201; expected 201; 572 ms
104 PASS	Approve manual intervention	HTTP 200; expected 200; 397 ms
105 PASS	Student practice includes intervention	HTTP 200; expected 200; 364 ms
106 PASS	Student cannot approve intervention	HTTP 403; expected 403; 251 ms
107 PASS	Student quiz list after attempt	HTTP 200; expected 200; 443 ms
108 PASS	Progress scores include completed quiz	HTTP 200; expected 200; 431 ms
109 PASS	Teacher gradebook includes completed quiz	HTTP 200; expected 200; 262 ms
110 PASS	Invalid PDF header rejected	HTTP 400; expected 400; 342 ms
111 PASS	Oversized upload rejected	HTTP 400; expected [400, 413]; 1534 ms
112 PASS	Reject task grade above maximum	HTTP 400; expected 400; 242 ms
113 PASS	Install valid PDF lesson fixture	HTTP 200; expected 200; 1226 ms
114 PASS	Dashboard schema diagnosis	missingColumns: ['quarter', 'description', 'is_active']; note: UI dashboard loads from other endpoints; the failed API is not its current data source.

# / Result	Check	Observed evidence
115 PASS	Taxonomy available to student	HTTP 200; expected 200; 564 ms
116 PASS	Create QA failing-score quiz	HTTP 201; expected 201; 510 ms
117 PASS	Read QA failing quiz	HTTP 200; expected 200; 562 ms
118 PASS	Current UI failing quiz submission	HTTP 201; expected 201; 1325 ms
119 PASS	Client score cannot override server grading	score: 0
120 PASS	Failing quiz triggers intervention	
121 PASS	Current UI duplicate assessment rejected	HTTP 409; expected 409; 513 ms
122 PASS	Analytics XLSX export	HTTP 200; expected 200; 358 ms
123 PASS	Analytics PDF export	HTTP 200; expected 200; 407 ms
124 FAIL	Assign uploaded lesson to intervention	HTTP 500; expected 201; 1676 ms; SQLITE_CONSTRAINT: SQLite error: FOREIGN KEY constraint failed
125 FAIL	Modify intervention with uploaded lesson	HTTP 500; expected 200; 382 ms; SQLITE_CONSTRAINT: SQLite error: FOREIGN KEY constraint failed
126 PASS	Override intervention completion	HTTP 200; expected 200; 460 ms
127 PASS	Peer cannot read student answer detail	HTTP 404; expected 404; 667 ms
128 PASS	Repeated missing-token rejection	HTTP 401; expected 401; 84 ms
129 PASS	Malformed bearer token rejected	HTTP 401; expected 401
130 PASS	Export populated XLSX report	HTTP 200; expected 200; 1888 ms
131 PASS	XLSX report contains assessment data	sheets: ['Master_Gradebook']; rowCount: 4
132 PASS	Export populated PDF report	HTTP 200; expected 200; 594 ms